Cloud Computing Course

Course n°2: Infrastructure as a Service

Maximilien Walbecq
Data Engineer
Roquette Frères 🌱

Quiz !

© Maximilien WALBECQ - Created with Marp e MarpX.

Quiz

1. What does on-premises mean ?

A. Applications and infrastructure fully managed by a cloud provider
B. Hardware and software hosted in the company’s own facilities
C. Data stored on portable devices only
D. Services accessed only through the internet

© Maximilien WALBECQ - Created with Marp e MarpX.

Quiz

1. What does on-premises mean ?

A. Applications and infrastructure fully managed by a cloud provider
B. Hardware and software hosted in the company’s own facilities
C. Data stored on portable devices only
D. Services accessed only through the internet

© Maximilien WALBECQ - Created with Marp e MarpX.

Quiz

2. What is virtualization ?

A. Running multiple physical servers as one virtual server
B. Moving all applications to the cloud
C. A backup technique for disaster recovery
D. The process of creating a virtual version of computing resources

© Maximilien WALBECQ - Created with Marp e MarpX.

Quiz

2. What is virtualization ?

A. Running multiple physical servers as one virtual server
B. Moving all applications to the cloud
C. A backup technique for disaster recovery
D. The process of creating a virtual version of computing resources

© Maximilien WALBECQ - Created with Marp e MarpX.

Quiz

3. What is Rapid Elasticity ?

A. Cloud services are accessible from any device
B. Resources scale up or down automatically based on demand
C. Usage is tracked and billed according to consumption
D. Resources are pooled and shared across multiple customers

© Maximilien WALBECQ - Created with Marp e MarpX.

Quiz

3. What is Rapid Elasticity ?

A. Cloud services are accessible from any device
B. Resources scale up or down automatically based on demand
C. Usage is tracked and billed according to consumption
D. Resources are pooled and shared across multiple customers

© Maximilien WALBECQ - Created with Marp e MarpX.

Quiz

4. What is Infrastructure as a Service ?

A. Virtualized compute resources (servers, storage, networking)
B. Ready-to-use apps over the internet (e.g., email, CRM)
C. Development & deployment environment with managed tools

© Maximilien WALBECQ - Created with Marp e MarpX.

Quiz

4. What is Infrastructure as a Service ?

A. Virtualized compute resources (servers, storage, networking)
B. Ready-to-use apps over the internet (e.g., email, CRM)
C. Development & deployment environment with managed tools

© Maximilien WALBECQ - Created with Marp e MarpX.

Quiz

5. What is Private Cloud ?

A. Dedicated infrastructure for one organization
B. Shared by organizations with common goals
C. Shared infrastructure managed by third-party providers
D. Combination of private & public clouds

© Maximilien WALBECQ - Created with Marp e MarpX.

Quiz

5. What is Private Cloud ?

A. Dedicated infrastructure for one organization
B. Shared by organizations with common goals
C. Shared infrastructure managed by third-party providers

© Maximilien WALBECQ - Created with Marp e MarpX.

Reminder of last session

Introduction to Cloud Computing

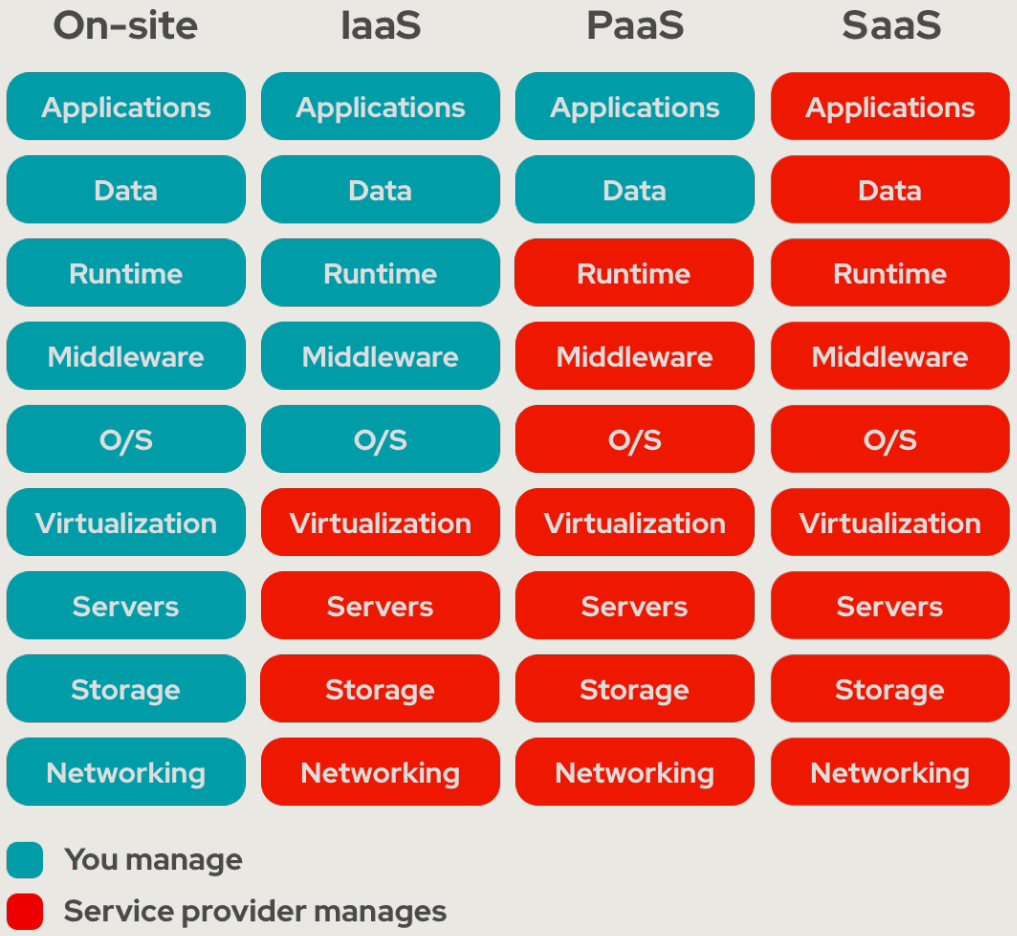
- Advantages of Cloud vs On-premise
- Data Center
- Virtualization
- Characteristics: Self-Service / Network Access / Elasticity / Measured / Resource Pooling
- Service: IaaS / PaaS / SaaS
- Deployment: Public / Private / Community / Hybrid

© Maximilien WALBECQ - Created with Marp e MarpX.

Reminder of last session

Infrastructure as a Service

- Access to fundamental computing resources: compute, storage, network
- Full control over deployed applications
- No physical server management



© Maximilien WALBECQ - Created with Marp e MarpX.

Additional notes for Azure Student Subscriptions

With Azure for Students you are restricted to five different regions, where you can deploy resources to, and these vary for each person.

To check:

- Go to -> Policy / Stratégie
- Authoring -> Assigments / Création -> Affectations
- Check "Allowed resource deployment regions"

My apologies for the inconvenience, I created an Azure for Student account to be sure that I will see the same as you.

© Maximilien WALBECQ - Created with Marp e MarpX.

Additional notes for Azure Student Subscriptions

For students without any licenses or credits on their account:

- Try to reclaim the subscription and recreate an account
- Work with a classmate

© Maximilien WALBECQ - Created with Marp e MarpX.

Contenu

Infrastructure as a Service

1. Title
2. Management Infrastructure, Identity & Access
3. Cloud resources: Storage
4. Cloud resources: Compute
5. Cloud resources: Network
6. How to define and deploy an Infrastructure on the Cloud

© Maximilien WALBECQ - Created with Marp e MarpX.

Aim of this course

The aim of this sessions is to:

- Understand IaaS fundamentals.
- Learn the basics of Infrastructure as Code and why it's important in cloud infrastructure management.
- Try to deploy an IT architecture through various tools

> **Warning**
> I have tried not to focus on Azure-specific resources in this course. However, it will remain centered on this cloud provider and make use of its terminology.

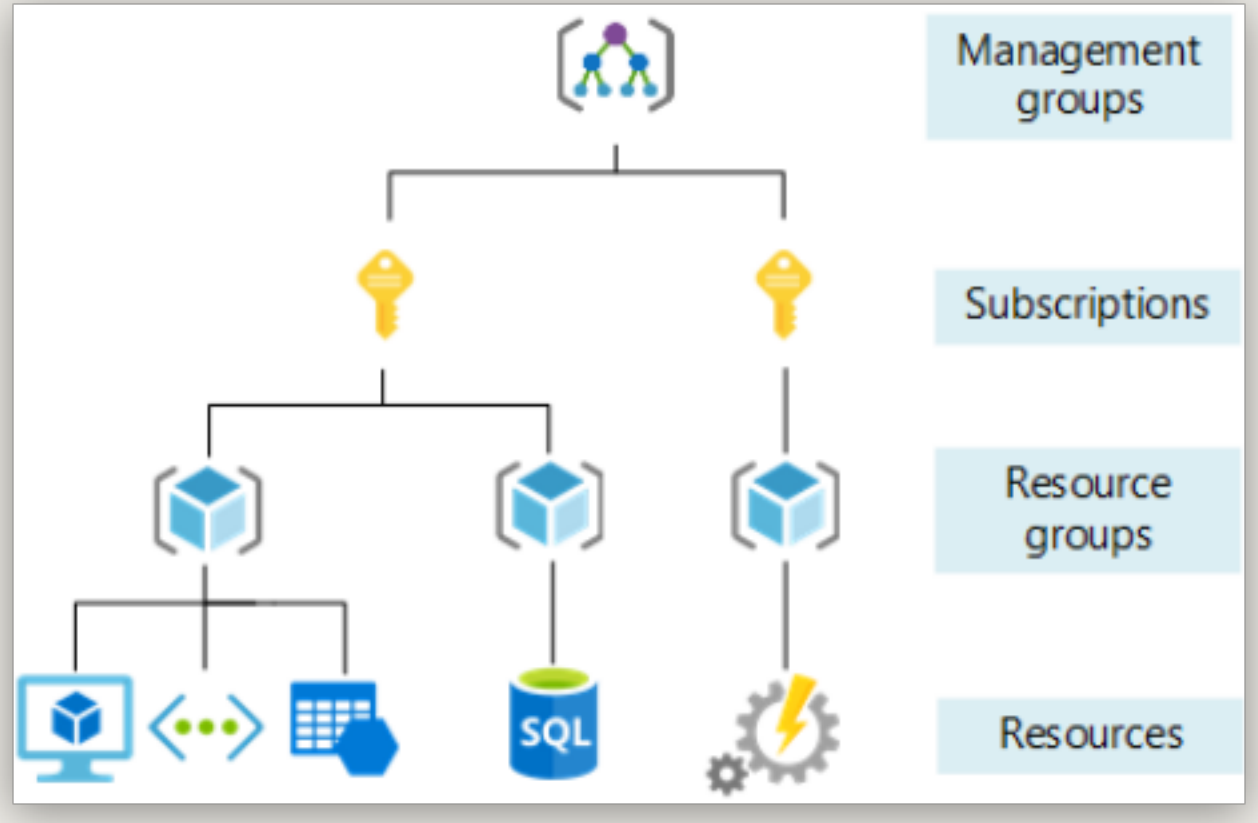
© Maximilien WALBECQ - Created with Marp e MarpX.

Cloud resources

Management Infrastructure, Identity & Access

The management infrastructure

- All cloud providers include a management layer for resources.
- Provides logical containers to group and organize resources.
- Enables access control, monitoring, and policy enforcement.
- Helps with cost tracking and resource lifecycle management.



© Maximilien WALBECQ - Created with Marp e MarpX.

Authentification vs Authorisation

Authentification : **who you are**

Authorisation : **what you are allowed to access to**

Permissions can be assigned to:

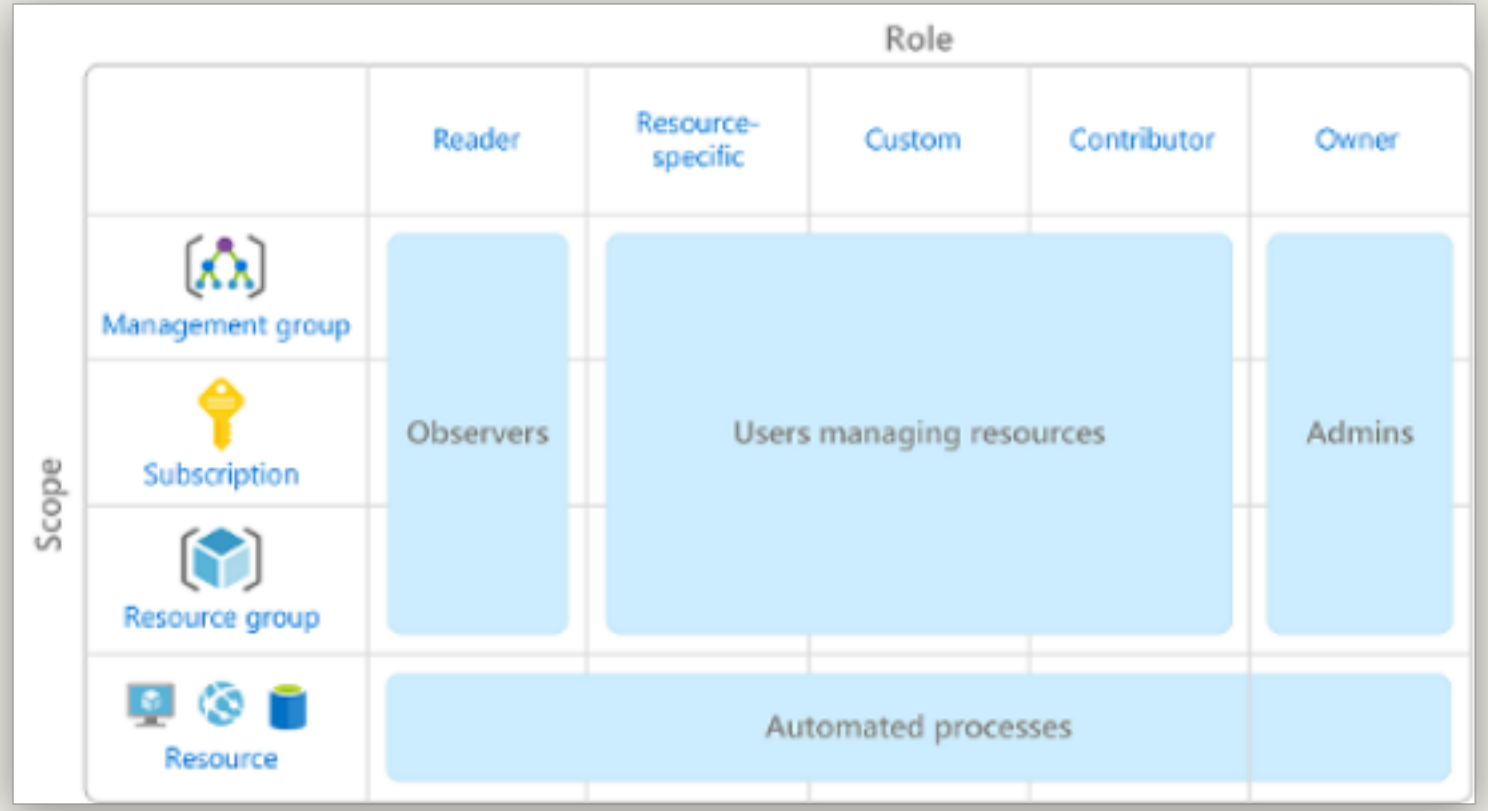
- User account: an individual identity with credentials
- Service Account: a cloud application or resource
- Group: a collection of users

© Maximilien WALBECQ - Created with Marp e MarpX.

Azure: Role-based access control (RBAC)

Warning

Apply the principle of least privilege when granting permissions to users or groups!



© Maximilien WALBECQ - Created with Marp e MarpX.

Cloud resources

Storage

Cloud resources: Storage Types

Multiple storage service on cloud:

- Blobs: A massively scalable object store for text and binary data.
- Files: Managed file shares for cloud or on-premises deployments.
- Queues: A messaging store for reliable messaging between application components.
- Block: Raw disk volumes for VMs.
- Tables: NoSQL table option for structured, non-relational data.

Storage service != Database service

© Maximilien WALBECQ - Created with Marp e MarpX.

Cloud resources: Storage Parameters

Tiers

- Hot: Optimized for storing data that is accessed frequently
- Cold: Optimized for storing data that is infrequently accessed
- Archive: Appropriate for data that is rarely accessed

Redundancy

Use redundancy to protect your data from transient hardware failures, network or power outages, and natural disasters.

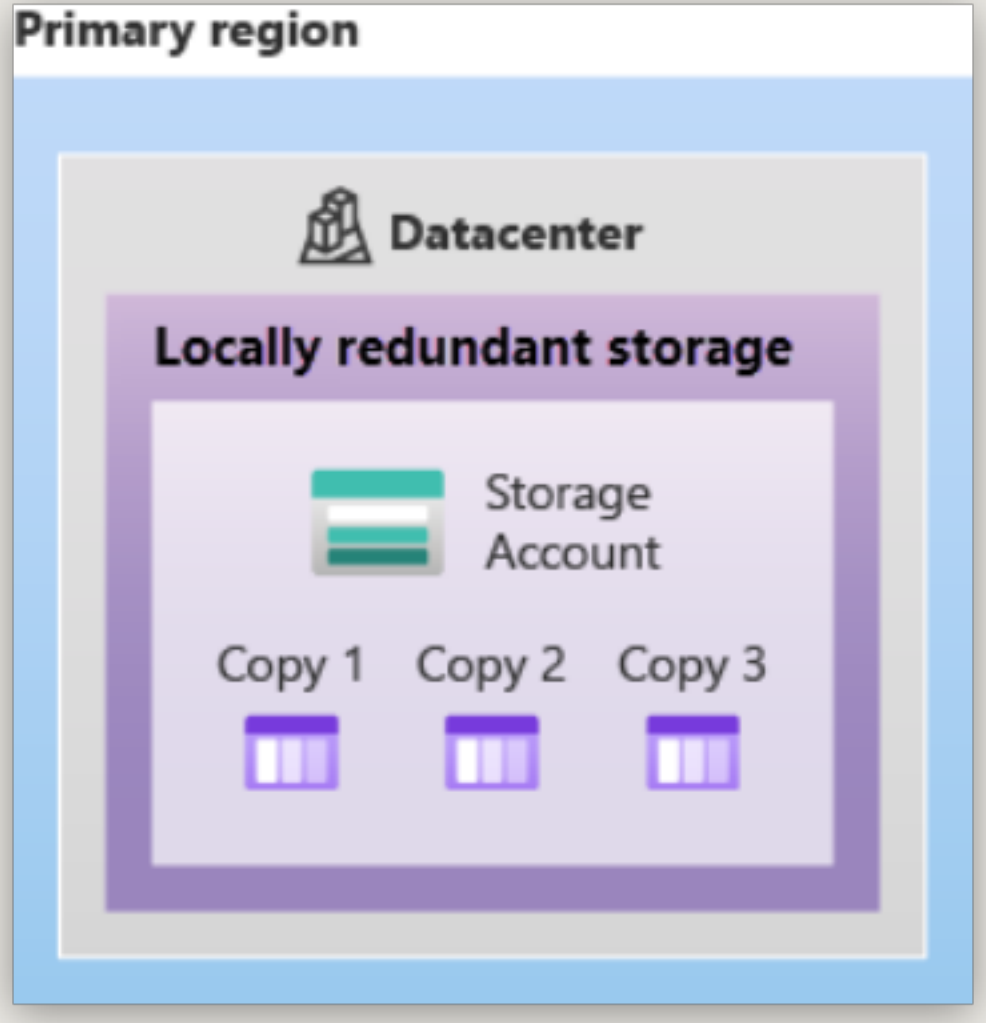
- Locally redundant storage (LRS)
- Zone-redundant storage (ZRS)
- Geo-zone-redundant storage (GRS)
- And many more ...

© Maximilien WALBECQ - Created with Marp e MarpX.

Cloud resources: Storage Redundancy

Locally redundant storage (LRS)

- Replicate data multiple times within a single data center
- Protect data against server rack and drive failures
- 11 nines of durability over a year (99.999999999%)



© Maximilien WALBECQ - Created with Marp e MarpX.

Cloud resources: Storage Redundancy

Zone-redundant storage (ZRS)

- Replicate data synchronously across multiple data centers in the same region
- Protect data against a data center failure (e.g. fire, flood)
- 12 nines of durability over a year (99.9999999999%)

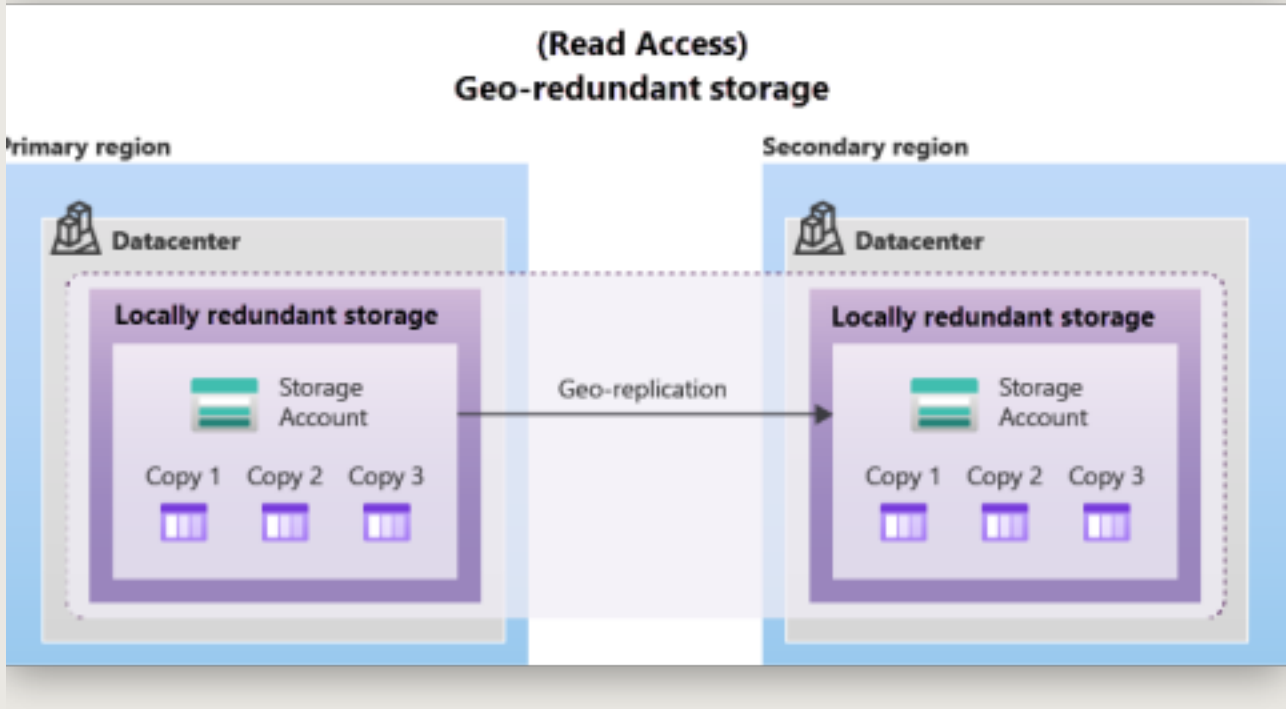


© Maximilien WALBECQ - Created with Marp e MarpX.

Cloud resources: Storage Redundancy

Geo-zone-redundant storage (GRS)

- LRS in a datacenter in primary region + LRS in another paired region
- Protect data against catastrophic events in primary region
- 16 nines of durability over a year (99.999999999999%)



© Maximilien WALBECQ - Created with Marp e MarpX.

Cloud resources

Compute

Cloud resources: Compute Types

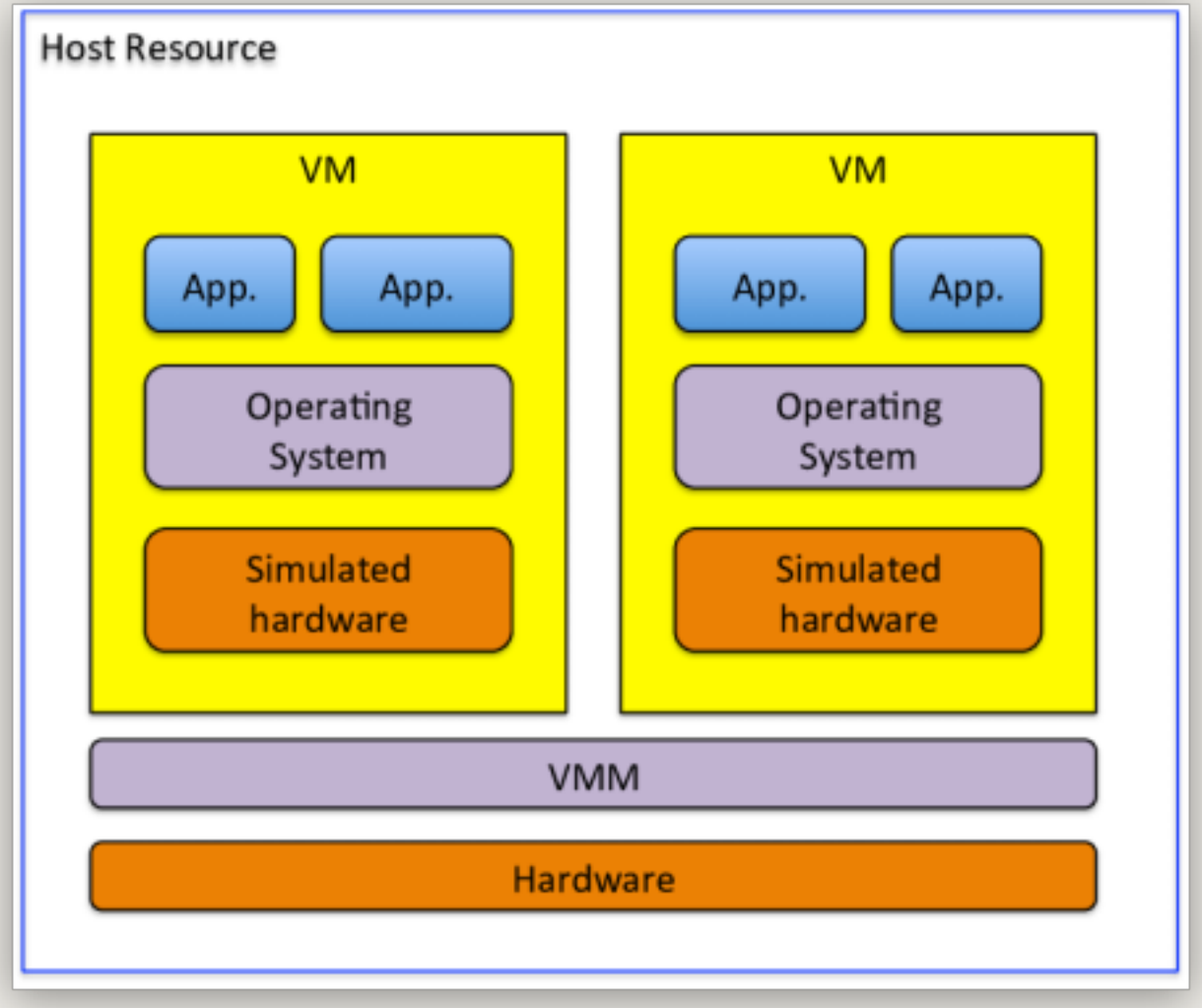
Definition:
Compute resources provide virtualized processing power to run applications and workloads in the cloud.

- Virtual Machines
- Containers
- Serverless Compute
- Specialized Compute

© Maximilien WALBECQ - Created with Marp e MarpX.

Compute: Virtual Machines

- **Cloud equivalent of a physical server**: runs an operating system and applications
- **Full control** over OS, CPU, memory, storage, and networking
- **Flexible scaling**: scale up (vertical) or add more instances (horizontal) as needed



© Maximilien WALBECQ - Created with Marp e MarpX.

Compute: Containers

- Lightweight, portable execution environments
- Encapsulates an application and its dependencies
- Optimized for microservices and modern applications
- Supports orchestration for automated scaling and management

© Maximilien WALBECQ - Created with Marp e MarpX.

Compute: Serverless Compute

- Event-driven execution model
- No server management, auto-scaling
- Pay only per execution

© Maximilien WALBECQ - Created with Marp e MarpX.

Compute : Specialized Compute

- High Performance Computing (HPC) for scientific simulations or analytics
- GPU / TPU workloads (AI, ML, rendering)
- Compute-optimized VM for batch processing and compute-heavy apps

© Maximilien WALBECQ - Created with Marp e MarpX.

Cloud resources

Network

Cloud resources: Network

Component	Purpose
Virtual Network (VNet)	Azure’s version of a private network (VPC)
Subnets	Segment networks for organization and security
Route Tables & Gateways	Direct traffic and connect networks (VPN, ExpressRoute)
Network Security Groups / Firewalls / ACLs	Control and filter inbound/outbound traffic
Load Balancers & Traffic Managers	Distribute traffic efficiently, enable high availability
VPN / ExpressRoute	Hybrid connectivity with on-premises networks

© Maximilien WALBECQ - Created with Marp e MarpX.

Cloud resources: Network

Objectives

- **Isolation & Segmentation**: Separate resources securely
- **Internet Communication**: Access Azure services or external endpoints
- **Resource-to-Resource Communication**: Within Azure, between VMs, storage, etc.
- **On-Premises Connectivity**: Connect cloud and local data centers
- **Traffic Routing**: Control the path of data flows
- **Traffic Filtering & Security**: Firewalls, NSGs, ACLs
- **Virtual Network Peering**: Connect multiple virtual networks

© Maximilien WALBECQ - Created with Marp e MarpX.

And many more



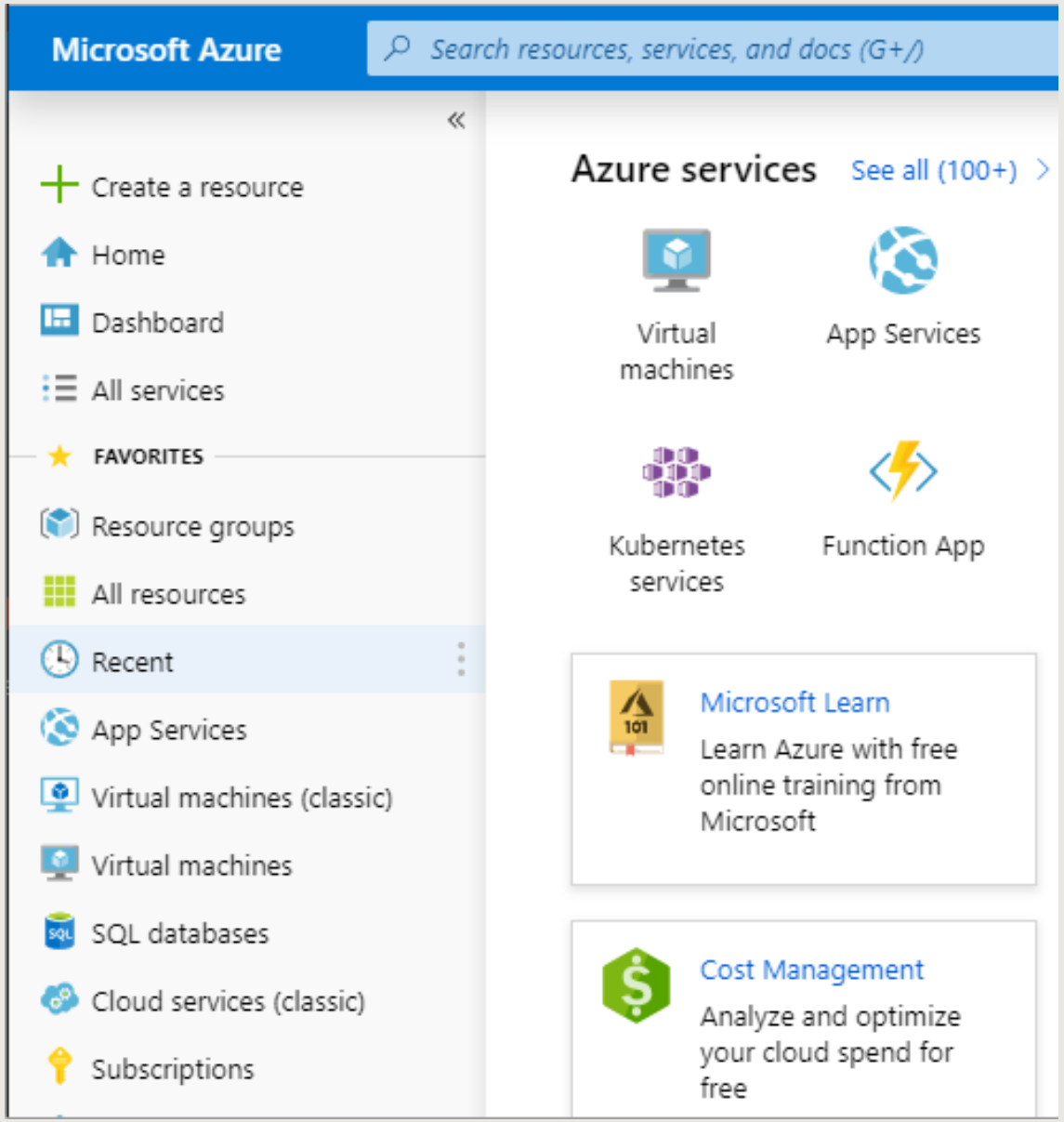
© Maximilien WALBECQ - Created with Marp e MarpX.

How to define and deploy

an Infrastructure on the Cloud

Azure Portal

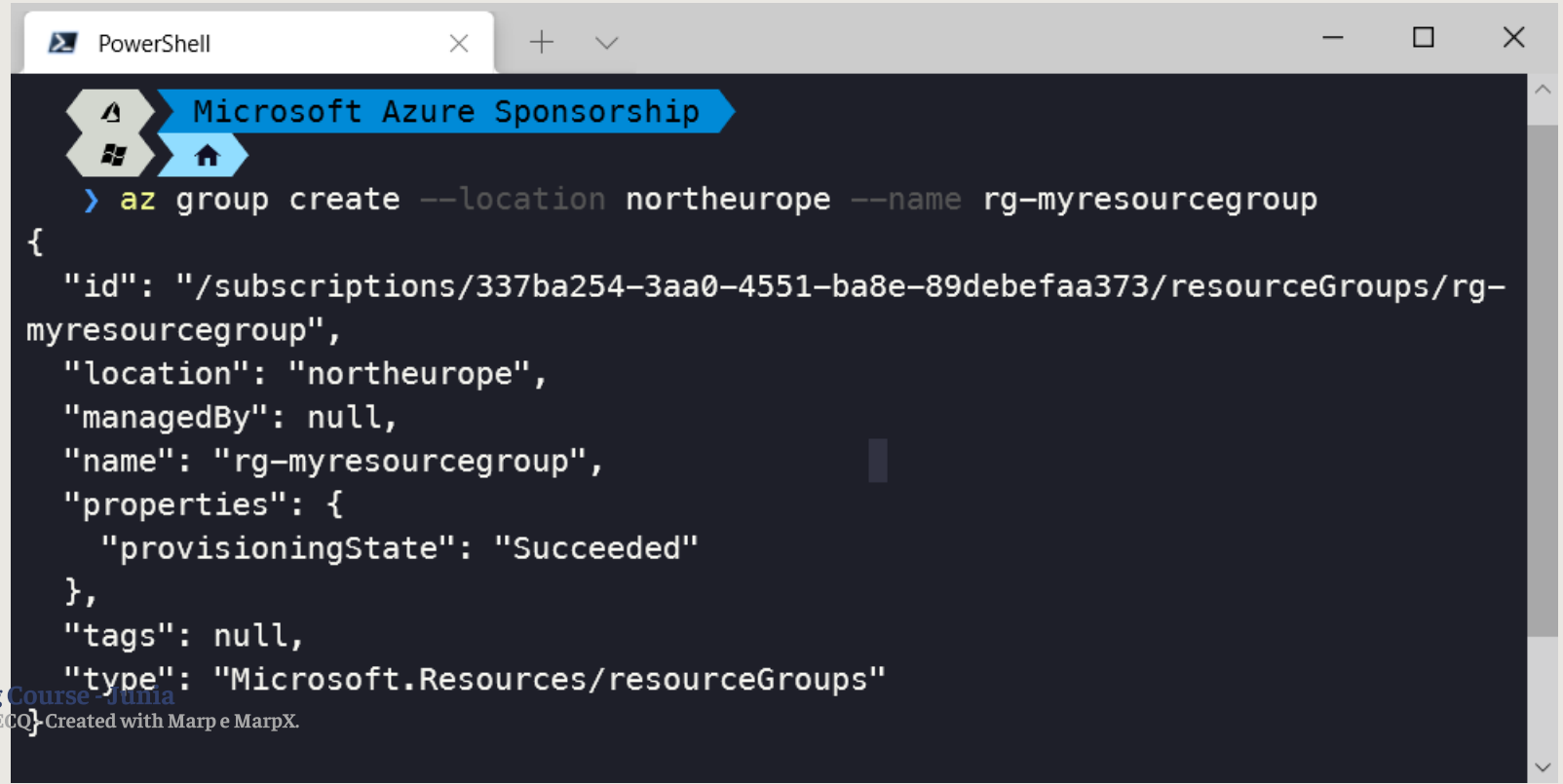
- **Visually intuitive**: easy to navigate and manage resources
- **Easy access**: web-based, available from anywhere
- **Prone to human errors**: manual configuration can lead to mistakes



© Maximilien WALBECQ - Created with Marp e MarpX.

Azure CLI

- **Azure-specific command-line tool**: manage only Azure resources via terminal
- **Automation-friendly**: ideal for repetitive tasks
- **Consistency & reproducibility**: reduces human errors compared to manual portal actions



© Maximilien WALBECQ} Created with Marp e MarpX.

Infrastructure as Code

Definition

Managing infrastructure using code instead of manual configuration.

Benefits

- Consistency & reproducibility
- Version control & collaboration
- Automation & efficiency
- Remove human error

© Maximilien WALBECQ - Created with Marp e MarpX.

Declarative versus Imperative

Declarative: Describe what the infrastructure should look like.

Benefits

- Simpler to read and understand
- Easier to write and define
- Simple way to ensure consistency across environments

Imperative: Describe how to achieve the infrastructure step by step.

Benefits

- More control over each step of deployment
- Can handle complex or custom operations easily
- Harder to manage

© Maximilien WALBECQ - Created with Marp e MarpX.

Your IaC should be idempotent!

- Applying the same IaC code multiple times produces the same result.
- Prevents accidental duplication or drift in infrastructure.
- Ensures consistency across environments (dev, staging, production).

© Maximilien WALBECQ - Created with Marp e MarpX.

Azure Resource Manager (ARM)

- **Azure-specific Infrastructure as Code (IaC) tool**: define and deploy Azure resources using JSON templates
- **Automation-friendly**: deploy multiple resources consistently in one operation
- **Consistency & reproducibility**: reduces human errors compared to manual provisioning
- **Version-controlled**: infrastructure definitions can be stored in Git and tracked over time

You can also use it with Azure CLI

JSON

```json
{
  "$schema": "https://schema.management.azure.com/sche
  "contentVersion": "1.0.0.0",
  "parameters": {
    "location": {
      "type": "string",
      "defaultValue": "[resourceGroup().location]"
    }
  },
  "resources": {
    "mystore": {
      "type": "Microsoft.Storage/storageAccounts",
      "apiVersion": "2023-04-01",
      "name": "mystorageaccount",
      "location": "[parameters('location')]",
      "sku": {
        "name": "Standard_LRS"
      },
      "kind": "StorageV2"
    }
  }
}
```


© Maximilien WALBECQ - Created with Marp e MarpX.

Terraform

- **Infrastructure as Code (IaC) tool**: automate deployment and management of cloud resources using declarative configuration files
- **Multi-cloud support**: works with Azure, AWS, GCP, and many others
- **Consistency & reproducibility**: reduces human errors compared to manual provisioning
- **Version-controlled**: infrastructure definitions can be stored in Git and tracked over time

main.tf

```
# Configure the Azure provider
terraform {
  required_providers {
    azurerm = {
      source  = "hashicorp/azurerm"
      version = "~> 3.0.2"
    }
  }

  required_version = ">= 1.1.0"
}

provider "azurerm" {
  features {}
}

resource "azurerm_resource_group" "rg" {
  name     = "myTFResourceGroup"
  location = "westus2"
}
```


© Maximilien WALBECQ - Created with Marp e MarpX.

Thank you !

Let's take a break before the practice session!